

Task 3: Communication Data, Signals, Events, and Protocol

1. **Data:**
Lead truck ID, obstacle status, braking status, deceleration, and distance between trucks.
2. **Signals:**
Obstacle detected, emergency brake warning, brake activation, and safe stop.
3. **Events:**
Obstacle appears, lead truck brakes, warning is sent, trailing truck receives it, trailing truck brakes, and both trucks stop safely.
4. **Sender and receiver:**
The lead truck sends the emergency brake warning to the trailing truck.
5. **Protocol:**
The scenario uses a V2V Emergency Brake Warning Protocol. When the lead truck detects the obstacle, it sends a warning message. The trailing truck receives it and starts emergency braking.
6. **Failure case:**
If the warning is not received, the trailing truck uses its own sensors and distance monitoring